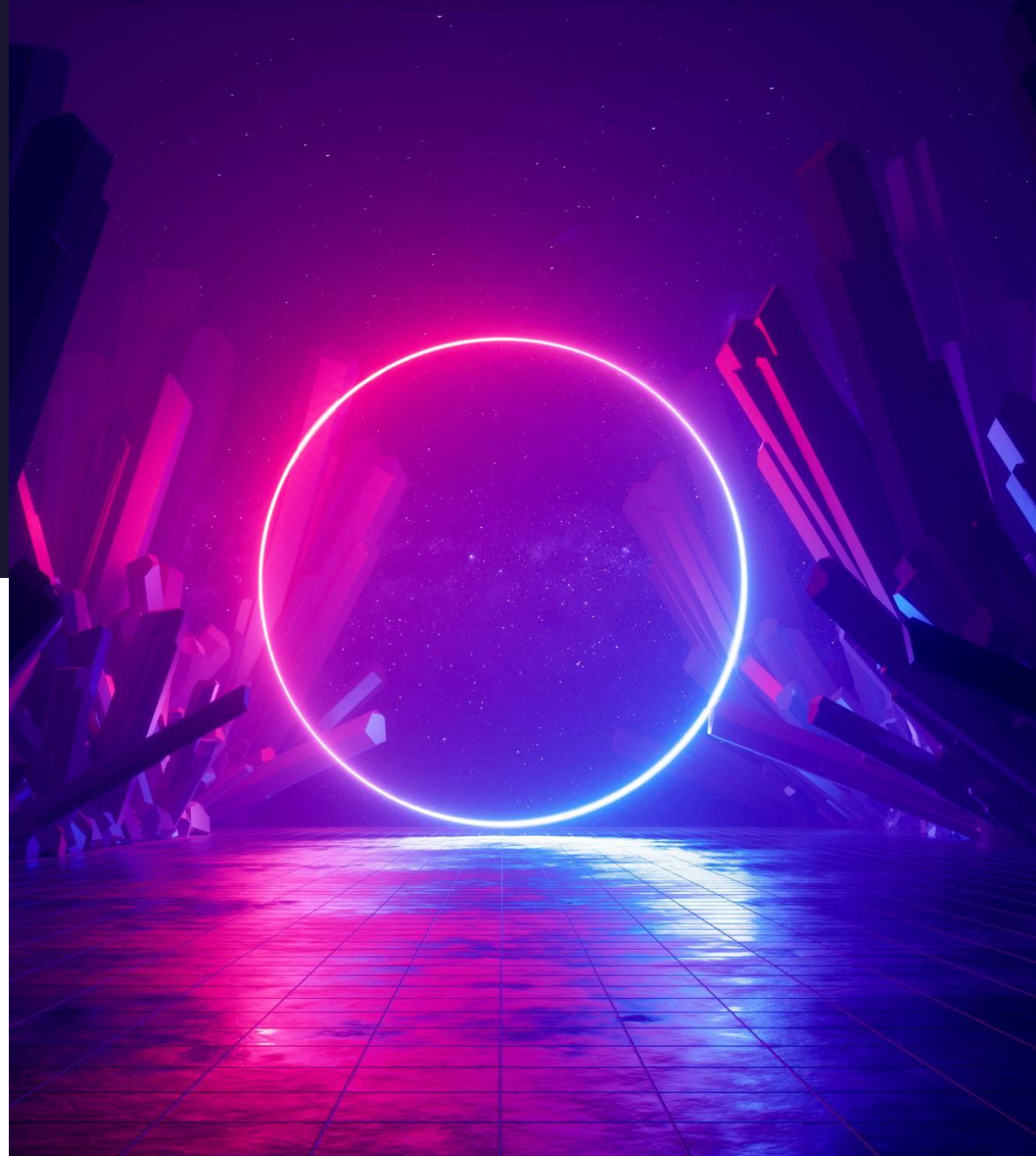
Electronics Engineering Technology

Profa. Dra. Talitha Nicoletti Regis



1° Semestre de 2023

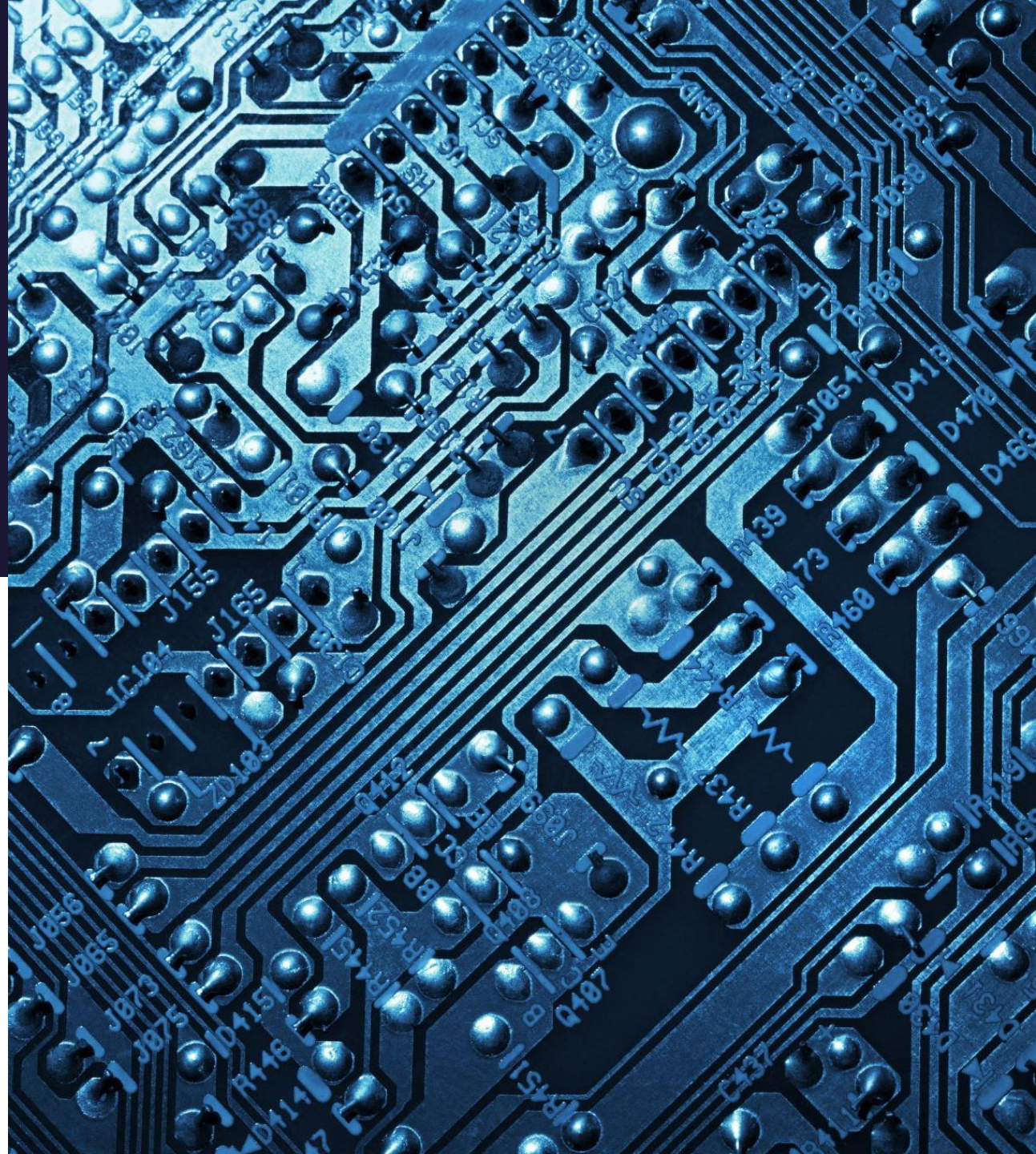
Disciplina: Electronics Engineering Technology



Ementa: Projeto e análise de circuitos eletrônicos com **diodos**, **transistores** e **amplificadores operacionais**. Utilização de componentes eletrônicos em circuitos práticos e no projeto de diferentes soluções, como dispositivos wearables e drones.

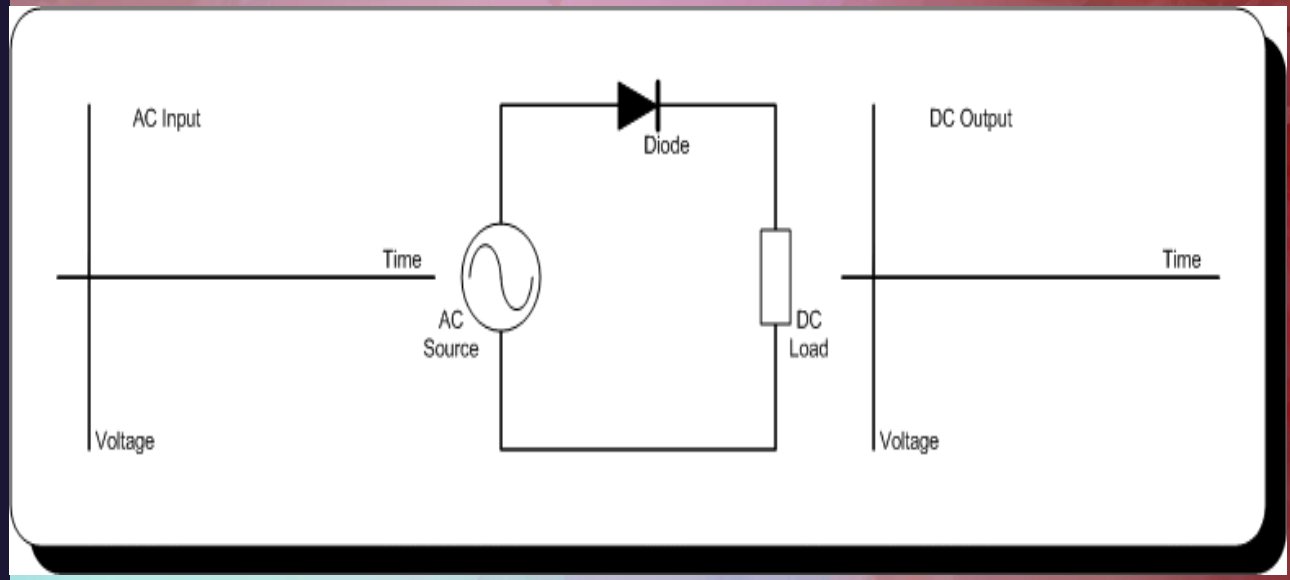
Conteúdo Programático

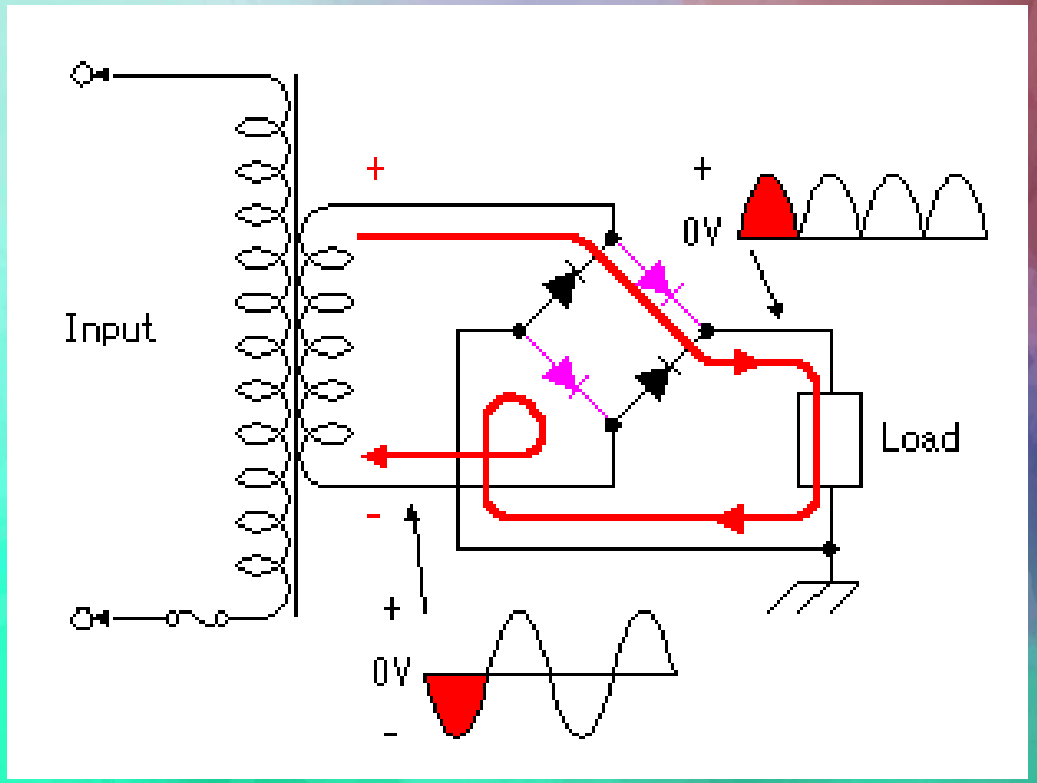
- ✓ Introdução à eletrônica
- ✓ Revisão
- ✓ Introdução a Semicondutores
- ✓ Introdução a Diodos
- ✓ Circuitos com Diodos
- ✓ Diodo Zener
- ✓ Diodo Emissor de Luz (LED)



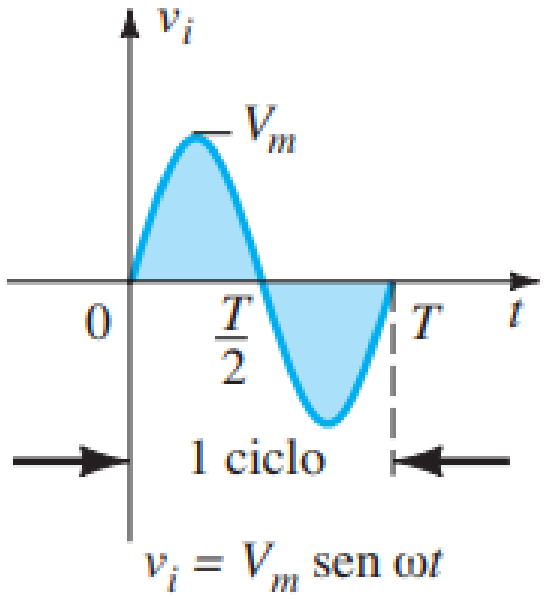
Revisão Circuitos Retificadores

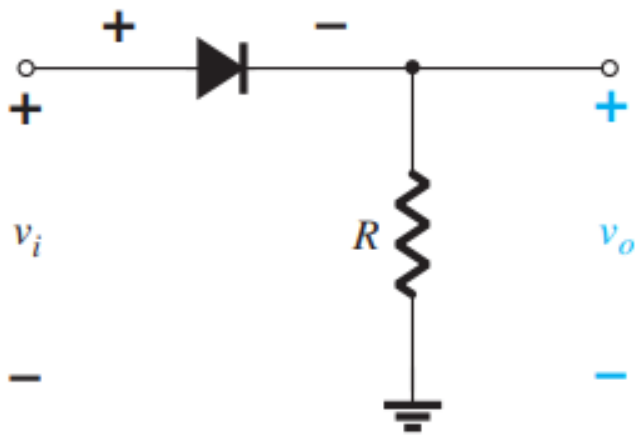
Meia onda e onda completa



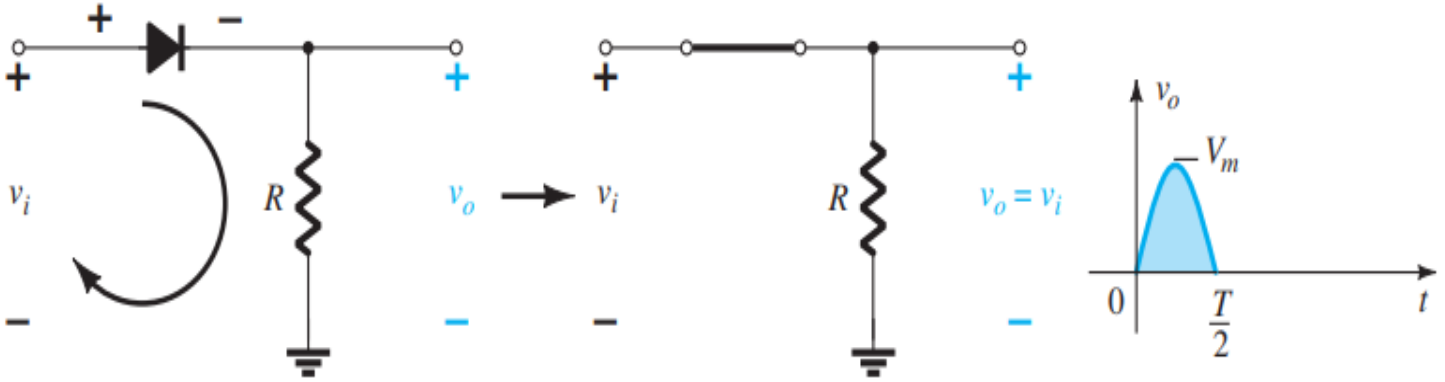


Retificação de Meia Onda

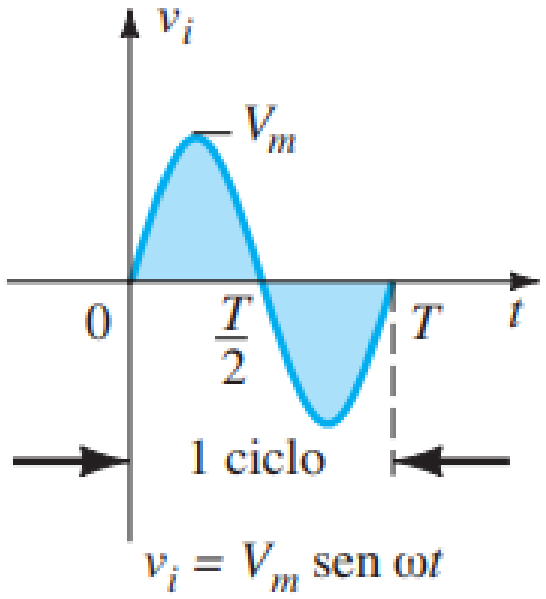


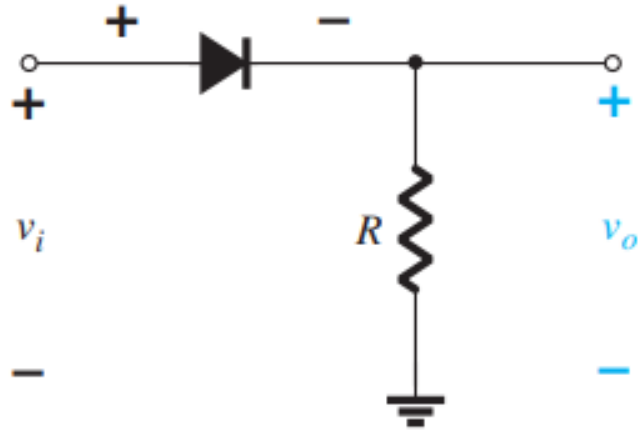


Região de condução ($0 \longrightarrow T/2$).

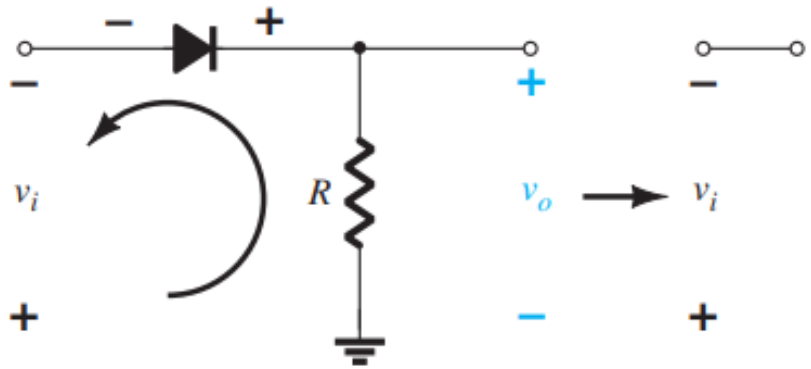


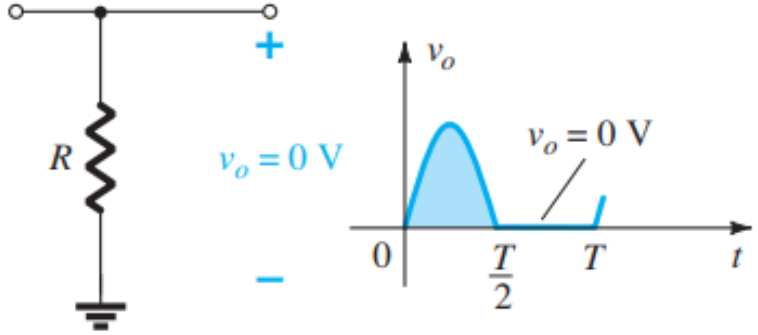
Retificação de Meia Onda



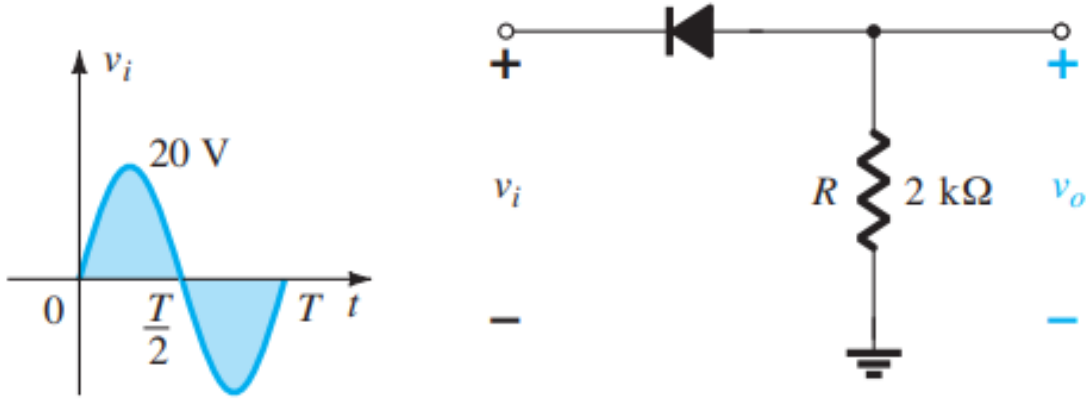


Região de não condução ($T/2 \rightarrow T$).





Exercício 1



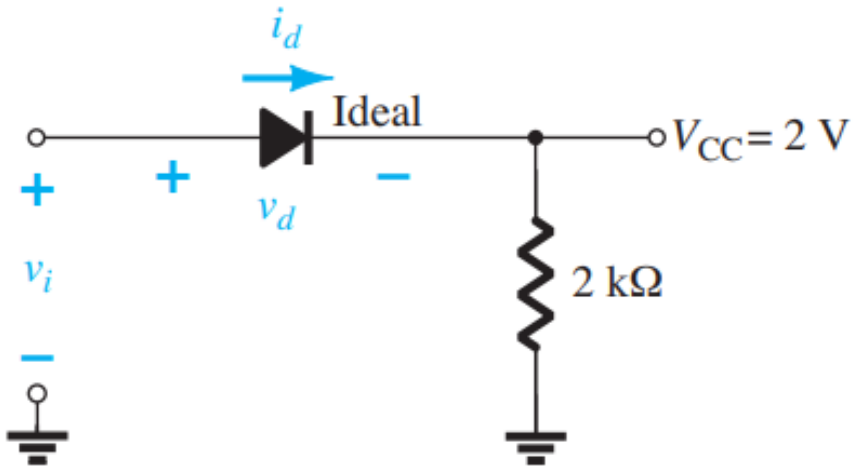
a)Esboce a tensão de saída vo e determine a tensão CC de saída para o circuito.

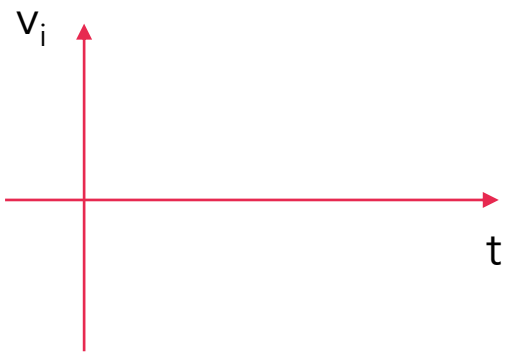
b)Repita se o diodo ideal for substituído por diodo de Si.

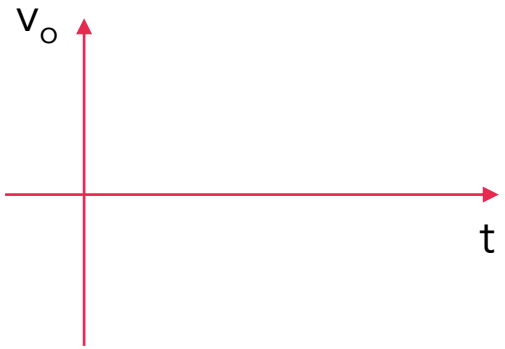
c)Se Vm=200V, qual será o valor de VCC, compare com o item anterior.

Exercício 2

Considerando um diodo ideal, esboce vi , vo e id para o retificador de meia-onda. A entrada é uma forma de onda senoidal com frequência de 60 Hz. Determine o valor de pico da entrada, os valores máximo e mínimo da tensão sobre o diodo e o valor máximo da corrente pelo diodo. Considere V_D=0,7V.

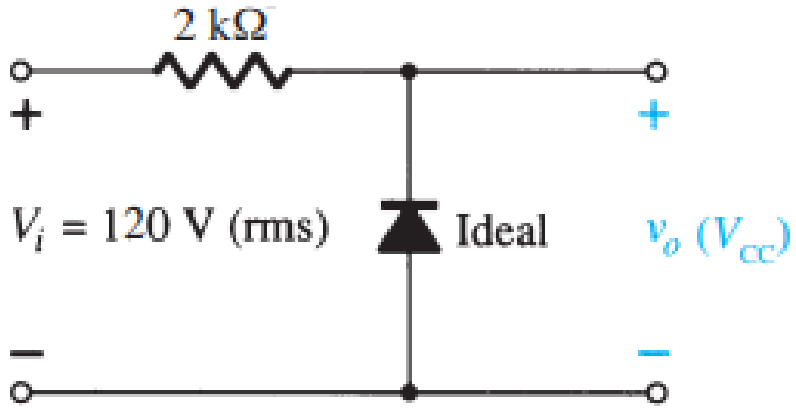


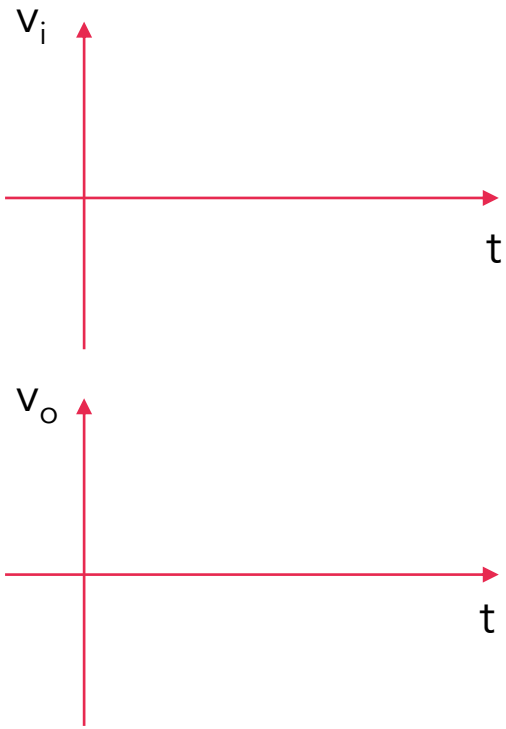




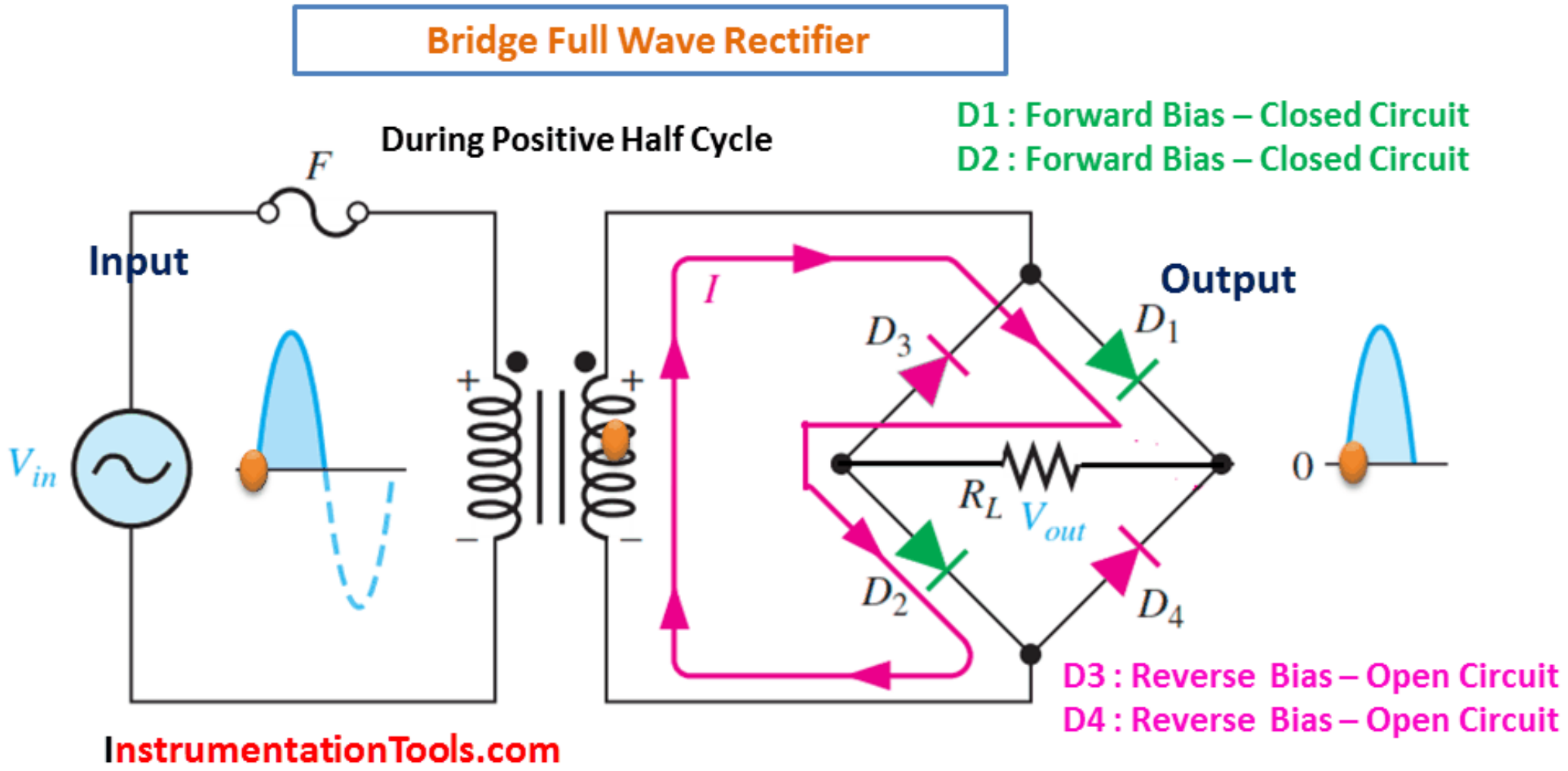
Exercício 3

Para o circuito esboce vo e determine VCC.

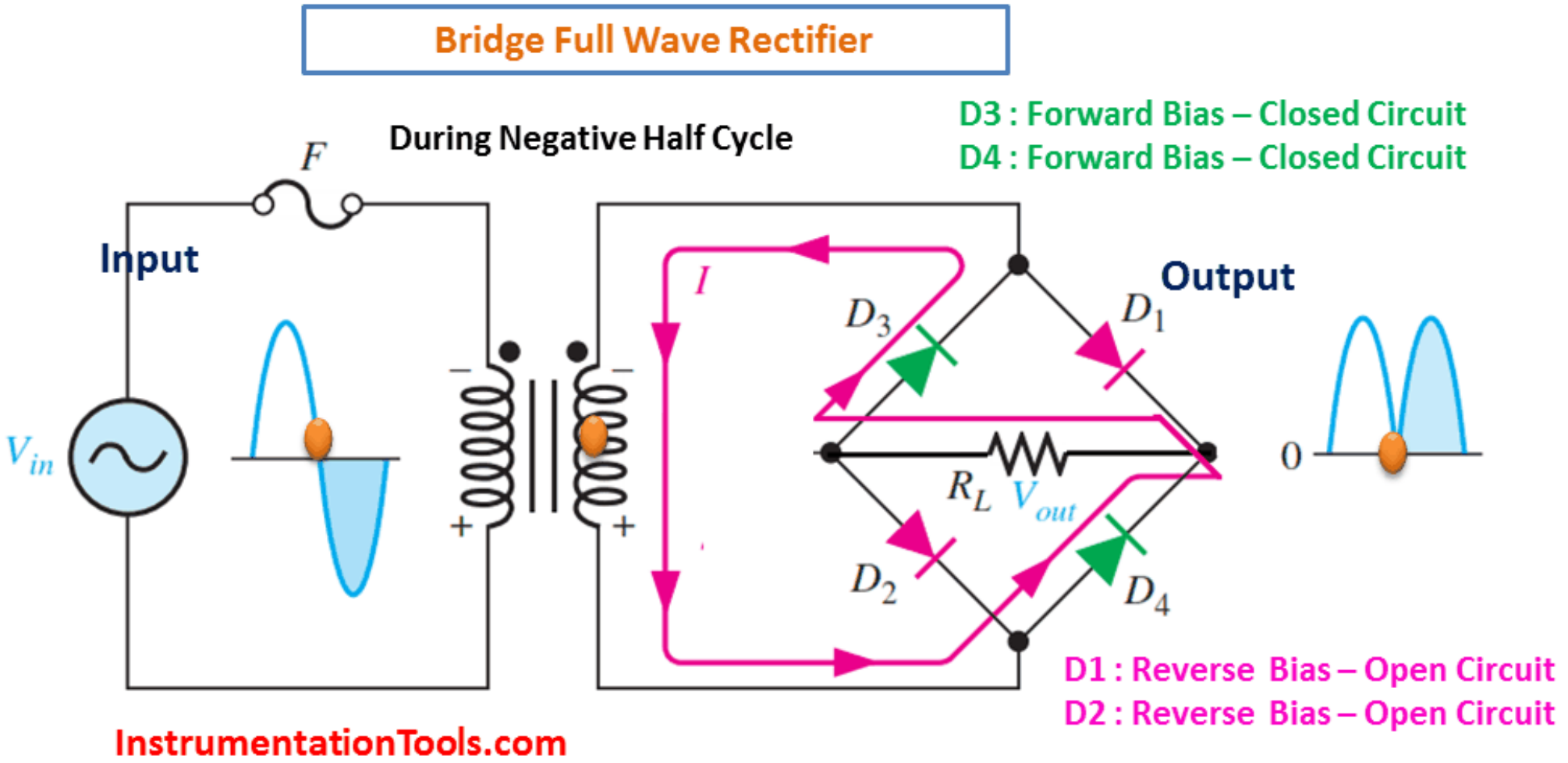




Retificação de Onda Completa

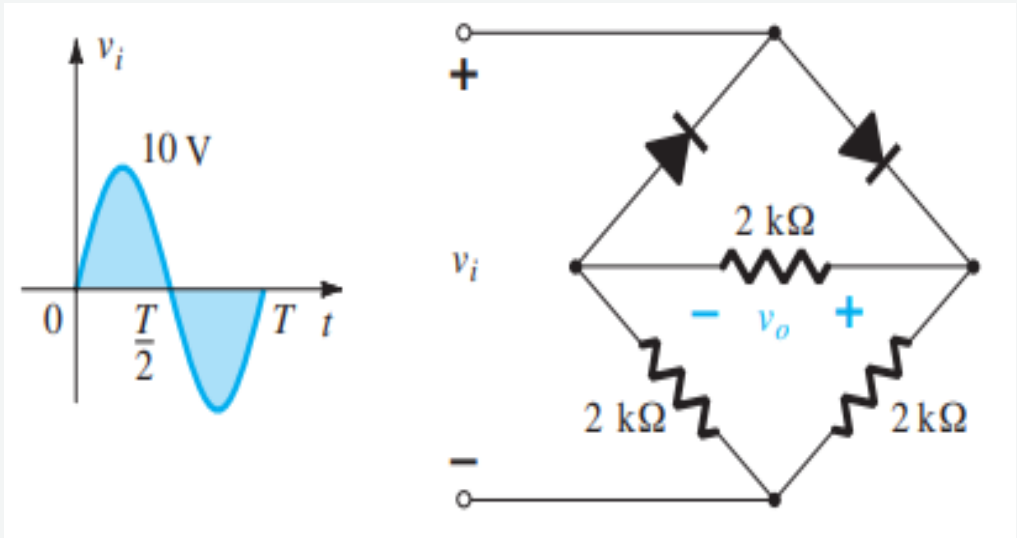


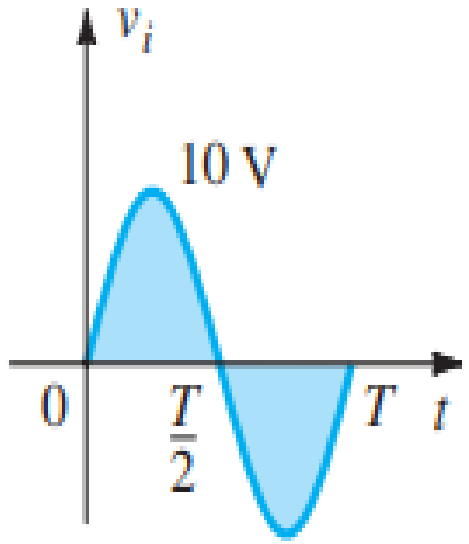
Retificação de Onda Completa

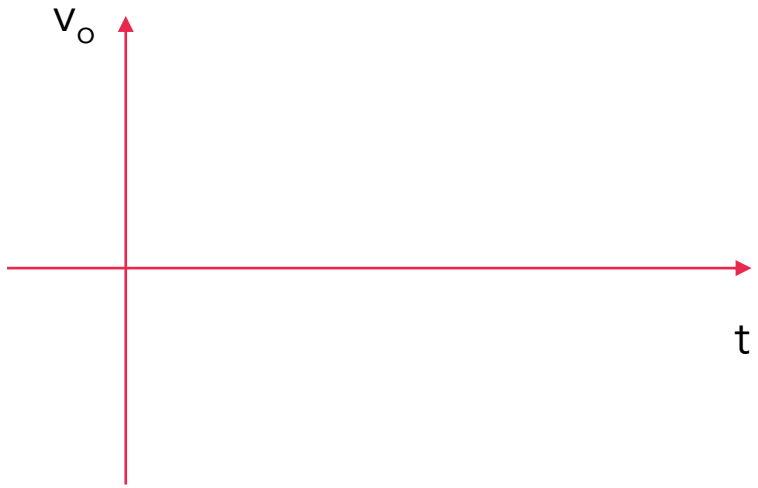


Exercício 4

Determine a forma de onda de saída do circuito, calcule o nível CC na saída e a PIV requerida para cada diodo.

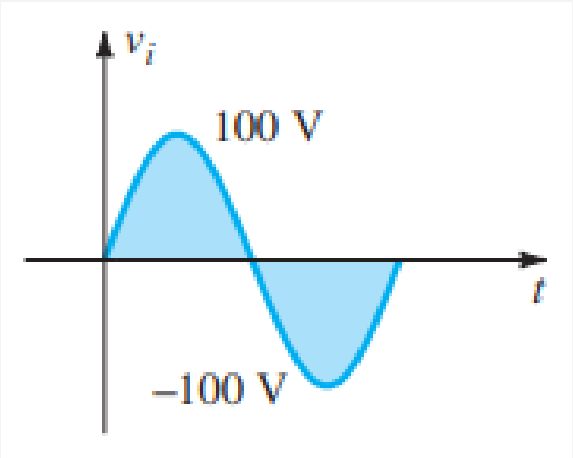


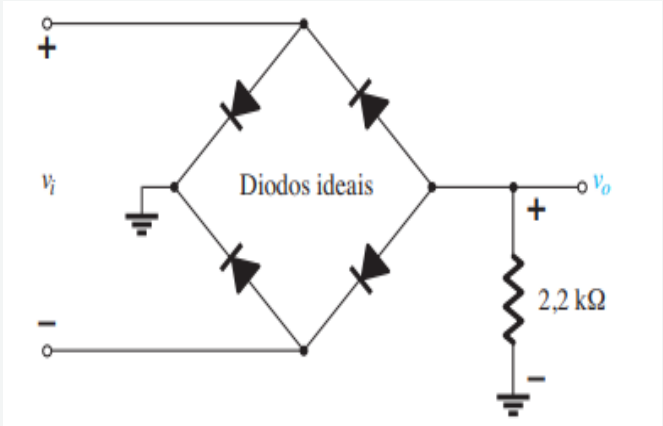


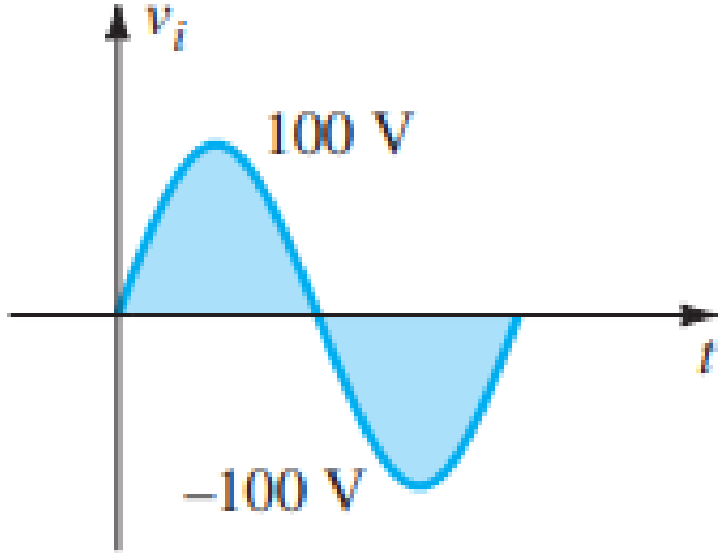


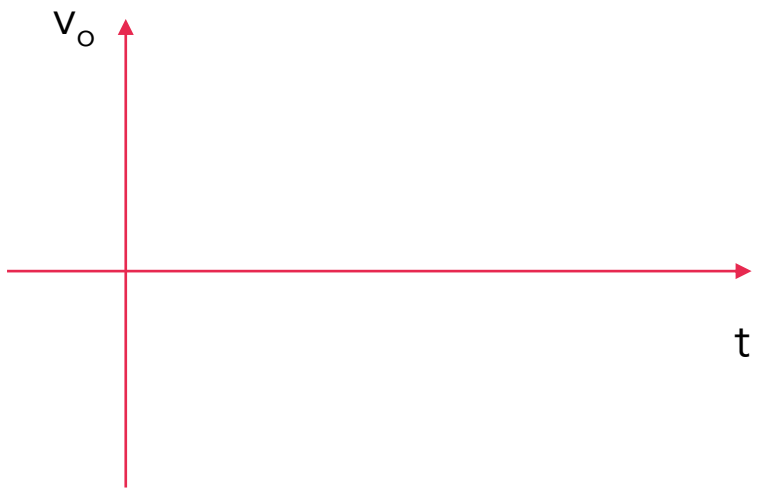
Exercício 5

Determine a forma de onda de saída do circuito, calcule o nível CC na saída e a PIV requerida para cada diodo. Determine também a corrente máxima através de cada diodo.









Exercício 6

Esboce vo para o circuito e determine a tensão CC disponível

